Odyssey

NATIONAL CYBER THREAT EARLY WARNING

FINAL ROUND · D3COY 1.0 · DJS ISACA · PROBLEM STATEMENT 01

See the *attack* coming.



Public threat intelligence in. For every Indian organisation: which systems are at risk, how sure we are, what to do, and by when. On one laptop, with no cloud.

SEVEN ASKS · ONE SYSTEM

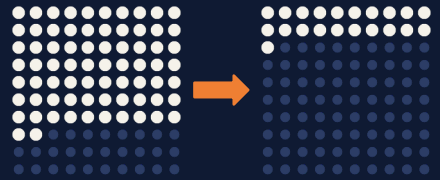
1 Collect · 2 Analyse · 3 Warn · 4 Prioritise · 5 Recommend · 6 Show · D Explain

BEHIND: THE LIVE THREAT MAP, AS THE CONSOLE DRAWS IT · RED = ATTACK SOURCE (DSHIELD, REAL) · VIOLET = COMPROMISE

The warning was public. The decision was *late.*

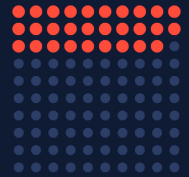
16,730

new CVEs published in October 2026 alone, from dozens of feeds. Every one needs a decision: does it touch us, how badly, how soon.



72% → 21%

of new CVEs say which product and version they affect, January against now



29%

of the CVEs exploited in 2026 were attacked on or before the day they were announced

one of 100 CVEs · labelled with product and version · attacked by day 0

REAL · CVE PROGRAM · NVD · CISA KEV

C-EDGE · 2024 · THE INDIAN CASE



31 Jul 2024 ransomware via the supplier; about 300 banks lose UPI and ATM

Jan 2024 Jenkins flaw disclosed; exploit public within days

19 Aug 2024 CISA lists the flaw, 19 days later

six months of public warning, nothing connecting it to who was exposed · event · the breach

SIEMENS WINCC · CISA ICS ADVISORY · REAL

One advisory, start to *finish*

1

Advisory arrives

CISA's ICS advisory for Siemens WinCC, read by the security LLM. No CVE id needed.

2

Versions parsed

Machine-readable ranges lifted from the text: below 8.0.0.11, and all versions.

3

Exposure found

13 WinCC assets at 7 operators, checked against the inventory version by version.

4

Risk assigned

A score with its working, an SSVC tier, a CERT-In clock. Patch, or mitigate where no fix exists.

5

People decide

A notice to each operator in its own language. An analyst approves. Every step audited.

THE SAME PATH FOR EVERY INPUT

a KEV entry | a new CVE from NVD | a decoy-grid surge | a partner's indicators | a SACHET alert

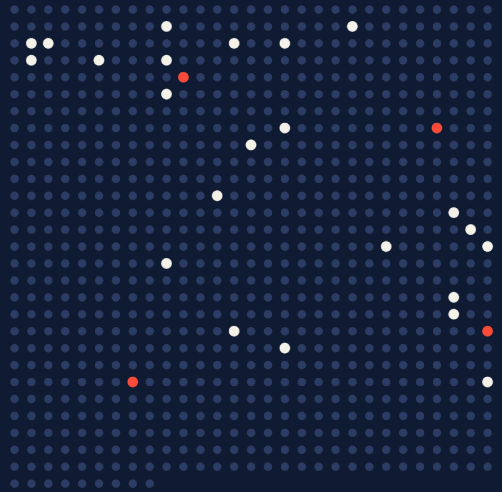
Each is matched to who is exposed, scored, and turned into something a person can approve.

○ automated, written to the audit trail ● the score ● a person decides

13 SOURCES · 10 REAL · 3 SIM

Every source, *one* record

MANIFEST	SOURCE
NVD 2021 to 2026	REAL
CISA KEV	REAL
CISA SSVC	REAL
FIRST EPSS	REAL
CISA ICS and alerts	REAL
MITRE ATT&CK	REAL
NDMA SACHET	REAL
DShield honeypots	REAL
Feodo C2	REAL
OpenPhish	REAL
CERT-In style advisories	SIM
Decoy grid, 24 sensors	SIM
Firewall flows	SIM



one dot = 100 CVEs published in 2026

names a product an operator runs

a Critical warning to a named operator

82,104

CVES PUBLISHED · 2026

2,277

NAME A PRODUCT WE RUN

354

CRITICAL, EACH TO A NAMED OPERATOR

In and out as STIX 2.1, TAXII 2.1 and CSV. A partner's indicators are matched against the record in seconds.

ALSO BUNDLED, REAL: EXPLOIT-DB · RANSOMWARE.LIVE · HIBP · GITHUB ADVISORIES · TOR · SPAMHAUS DROP · ET

9 MODELS · ONE RTX 4070 · 0 CLOUD CALLS

AI explains. *Rules* decide.



INTEL › GRAPH · 115 NODES · 447 LINKS

colour = technology family

compromise

threat group, overlap not attribution

live signal

22/23 — **Reads**

MiST 8B, a security LLM, reads every advisory and writes every brief. 22 of 23 briefs pass the fact validator; a failing sentence is withheld, never shown.

12 — **Relates**

SecureBERT 2.0 links entities and threats; a clustering model groups 12 active campaigns. Threat groups are ATT&CK overlap, never attribution.

5/5 — **Detects**

Surges, C2 beacons and flow anomalies, every 5 seconds. A poisoned advisory with hidden instructions: caught by the DeBERTa guard, 5 of 5.

MIST 8B · FOUNDATION-SEC 8B · SECUREBERT 2.0 · TF-IDF CLASSIFIERS · DAY-0 FORECAST · BAYES · LIVE DETECTORS · DEBERTA GUARD · INDICTRANS2, 22 LANGUAGES

No model sets a score, a tier or a deadline. One model per job, fixed in config, chosen by measurement on 2026 CVEs, newer than any model's training data.

Move the line *left*

+20.5 days

MEDIAN LEAD OVER CISA KEV · REAL SIGNALS · UP TO 42.4

THEN · REACT AFTER THE BREACH

Flaw disclosed
public record; exploit code soon after

Nothing connects it
to who actually runs the product

Breach
the warning signs were public for months

Listed
the national list catches up weeks later

NOW · ACT ON THE SIGNAL

12 days before
a decoy-grid surge (SIM) raises a warning

Day 0
matched to installed versions in seconds

Warning
tier, confidence, clock; a person approves

+20.5 days
median before CISA lists it, on real signals

61%

of later-exploited CVEs flagged on day 0 at a 2% alert budget; CVSS ≥ 9 catches 17% at the same budget

1 tick

one 5-second tick to catch a scanning surge or an implant beacon in the drills; 0 alerts on clean traffic

15 min

between feed reads; a new NVD CVE naming our products is matched, scored and forecast within seconds

reactive step · Odyssey step · the breach · the decision, moved before it

SIM = simulated sensor, labelled · numbers recomputed live on the Proof page

SCORING.JSON · HASHED INTO EVERY ALERT

A score you can check *by hand*

Risk = 100 ×

RELEVANCE ×	IMPORTANCE ×	LIKELIHOOD	SEVERITY	EVIDENCE
0 → 1	0.4 → 1.3	55%	30%	15%
this exact version affected?	criticality · exposure · blast radius	from evidence, in rank order	SSVC technical impact	Admiralty-graded sources

LIKELIHOOD · FROM EVIDENCE · IN RANK ORDER

On CISA KEV · SSVC says active · exploit seen on 2+ decoys	1.0
The vendor says exploited in the wild	0.8
A public proof of concept	0.5
A scanning surge on the decoy grid, decaying with age	0.4
EPSS ≥ 0.5 · CVSS ≥ 9 · automatable	corroborate
The day-0 forecast	prediction

Tier by CISA's SSVC decision path: Act, Attend, Track. **Confidence** scored apart: A to F source grades; agreement between independent sources raises it. **Relevance** checked against NVD, the vendor and the advisory: 0 false "affected" on the Ivanti case.

ONE REAL WARNING · TAPTI THERMAL · IVANTI CS 22.8R2.4

Likelihood 0.55 × 1.0 — 45.8
KEV 30 Oct · SSVC active · vendor: in the wild

Severity 0.30 × 1.0 — 25.0
SSVC technical impact: total

Evidence 0.15 × 1.0 — 12.5
CISA (A), NVD (A), vendor (B) agree

Relevance × 1.0 · **Importance** × 0.83
affected version · VPN gateway, internet-facing, tier 1

RISK — 83.3
Critical · Act · 12 h to patch · P(compromise) 18%

A PERSON APPROVES ANYTHING CONSEQUENTIAL

What to do, and *by when*

ON EVERY WARNING

01 **Patch** to a named fixed release, never a downgrade

02 **Block** the indicators: STIX 2.1 export, risk lists for the firewall, C2 refanged

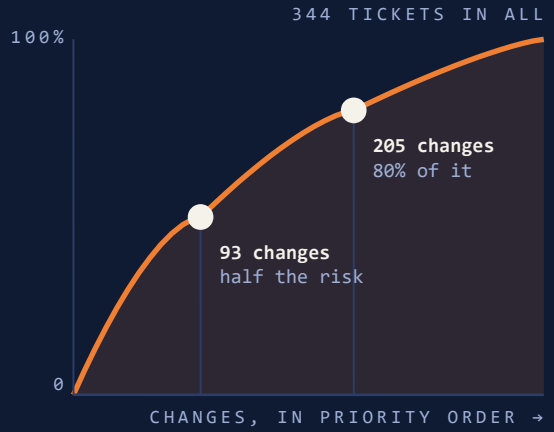
03 **Investigate** a hunt playbook, ATT&CK mitigations, forensic triage where CISA asks

04 **Report** the CERT-In 6-hour report pre-drafted; RBI, SEBI, DPDP and NCIIPC clocks shown

05 **Notify** a notice in the operator's own language; a person approves before it goes

Odyssey proposes; a person approves anything consequential. Every action is in a hash-chained audit trail.

THE RISK PLAN



BY WHEN · CERT-IN

6 h suspected compromise

12 h known-exploited, internet

24 h Critical

72 h High

120 h Medium

DIAL = DEADLINE ON A 120-HOUR FACE

○ runs on its own, audited ● a person decides one workflow per warning: issued → acknowledged → mitigating → verifying → closed

One screen per *question*

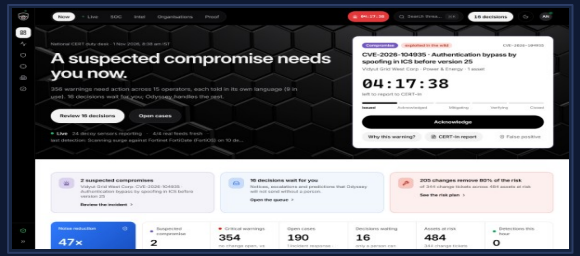


FIG 9A · NOW · WHAT NEEDS ME

chip = compromise clock = CERT-In deadline

black button = the one action · operators SIM

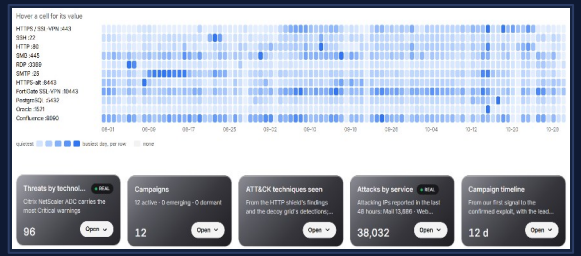


FIG 9B · INTEL · WHAT IS UNDER ATTACK

cell = attacks on a port that day, quiet → busy

DShield, REAL · rows = the services operators expose

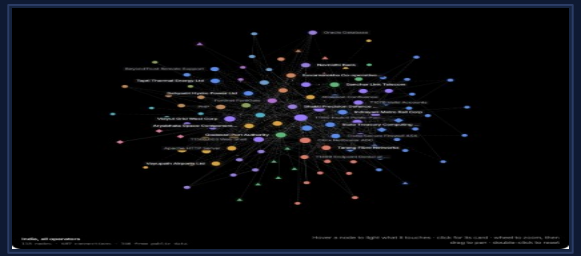


FIG 9C · INTEL · THE GRAPH

colour = technology family compromise

threat group (overlap) live signal

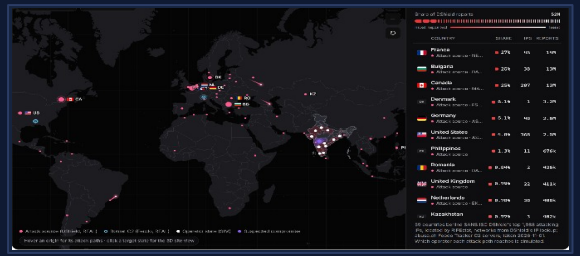


FIG 9D · NOW · WHERE ATTACKS START

attack source, REAL botnet C2, REAL

operator state, SIM compromise

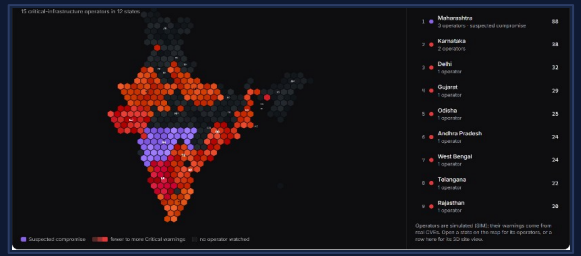


FIG 9E · INDIA BY WARNINGS

Critical warnings per state, few → many

compromise operators SIM, warnings from REAL CVEs



FIG 9F · A STATE, IN 3D

agency site, SIM attack from a real origin

origins REAL · buildings SIM

EVERY NUMBER RECOMPUTED LIVE ON THE PROOF PAGE

It runs here. *Offline.*

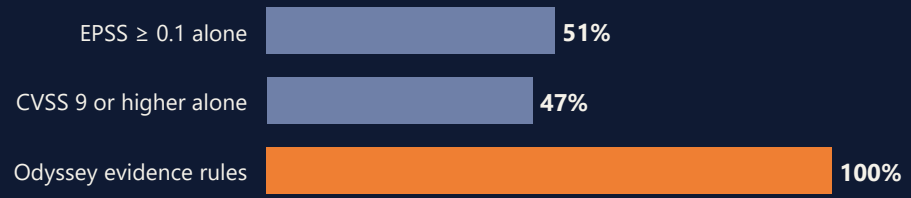
0
cloud or internet AI calls, ever

1
laptop, an RTX 4070 with 8 GB; every model under 8B

85 s
to re-score the nation: 15 operators, 612 systems, 2,277 flaws

221
engine tests green; 26 of 26 product flows; 25 of 26 browser steps

110 real exploited CVEs (CISA KEV, Aug to Oct 2026): share each rule would have flagged



KEV is one of our inputs, so 110 of 110 is by design. The honest test is out of time: the day-0 forecast, frozen in July, caught 61.1% of the CVEs exploited from August to October while flagging 2%; CVSS ≥ 9 at the same budget caught 16.7%.

- **Real, public**
  CVEs, KEV, SSVC, EPSS, CISA advisories, DShield, Feodo, OpenPhish, SACHET, Exploit-DB, ransomware.live, breach and network lists.
- **Simulated, labelled**
  the 15 operators and their inventory, the decoy grid, the firewall logs.
- **Proven on the wire**
  45 of 45 real web attacks detected and blocked inline, 0 false positives, about 15 ms each.

Real data. Measured claims. A human in the loop. *Zero cloud calls.*